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Dynamical Systems - MA3081

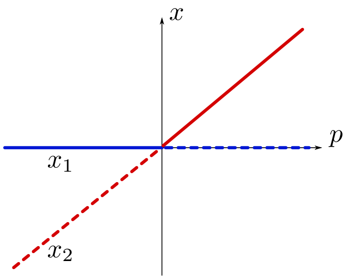
Solutions to Exercise Sheet 7

Exercise 7.1 (Transcritical bifurcation)**.**
Consider the one-dimensional dynamical system induced by

$$x' = x(p - x) \tag{7.1}$$

where $p \in \mathbb{R}$ is a parameter. At $p = 0$ a transcritical bifurcation occurs, i.e. the two equilibria $x_1 = 0$ and $x_2 = p$ exchange their stability, see also Example 8.9. Let $\varepsilon > 0$ and perturb the vector field of (7.1) by $\pm\varepsilon$. Analyze the bifurcation behavior of the resulting systems.

Solution. Recall the bifurcation behaviour of a transcritical bifurcation in normal form $x' = x(p - x)$: We have two equilibria at $x_1 = 0$ and $x_2 = p$. We see that x_1 is stable for $p < 0$ and unstable for $p > 0$, for x_2 we get the reversed behavior. Hence, for $p = 0$ the two equilibria exchange their stability. The bifurcation diagram looks as follows:



bold line: stable equilibrium

dashed line: unstable equilibrium

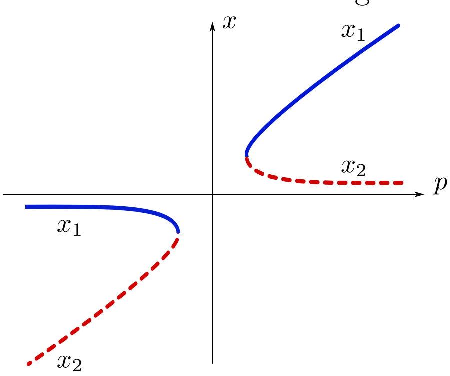
Let us first consider a perturbation by $-\varepsilon$, i.e. we have , $x' = x(p - x) - \varepsilon =: g(x, p)$:
The equilibria of this system are given by

$$x_{1,2} = \frac{p}{2} \pm \sqrt{\left(\frac{p}{2}\right)^2 - \varepsilon}.$$

Note that for $|p| < 2\sqrt{\varepsilon}$ no equilibrium exists. The stability of the equilibria for $|p| > 2\sqrt{\varepsilon}$ is again determined by analyzing the linearized system. $\partial_x g(x, p) = p - 2x$, i.e. we get:

$$\partial_x g(x_{1,2}, p) = p - p \mp 2\sqrt{\left(\frac{p}{2}\right)^2 - \varepsilon},$$

hence x_1 is stable and x_2 is unstable. The bifurcation diagram looks as follows:



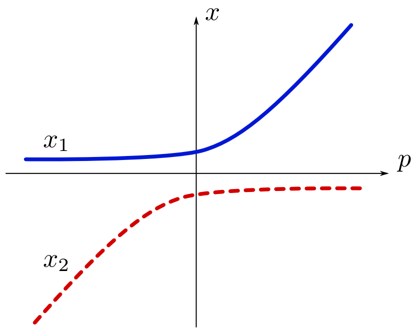
We observe that the transcritical bifurcation is split into two fold bifurcations at $p = \pm 2\sqrt{\varepsilon}$.
Now, let us analyze the behavior when we perturb (1) by the term $+\varepsilon$, i.e. we have $x' = x(p-x) + \varepsilon =: h(x,p)$.
The two equilibria

$$x_{1,2} = \frac{p}{2} \pm \sqrt{\left(\frac{p}{2}\right)^2 + \varepsilon}$$

exist for all values of p and since

$$\partial_x h(x_{1,2}, p) = p - p \mp 2\sqrt{\left(\frac{p}{2}\right)^2 + \varepsilon},$$

x_1 is stable and x_2 is unstable for all values of p. Hence, there are two non intersecting equilibrium branches, one stable and one unstable and no bifurcation occurs as p is varied. The bifurcation diagram looks as follows:



Remark: Note that finding explicit representations of the equilibria in terms of p and their stability is quite elaborate. Furthermore, drawing the corresponding curves in the bifurcation diagram is not easily done by hand. As an alternative one may solve the equilibrium equation $x' = ... = 0$ for p instead of x (after ruling out p-independent solutions). For example in the case of $x(p-x) - \varepsilon = 0$ we obtain $p = \frac{\varepsilon}{x} + x$, so we just have to draw the corresponding graph over the x-axis. □

Exercise 7.2 (Transcritical bifurcation II).
Consider the dynamical system induced by

$$x' = r \ln x + x - 1,$$

where $r \in \mathbb{R}$ is a parameter. Show that for a certain value of r a transcritical bifurcation occurs. Find new variables X and R that bring the system into the approximate normal form of a transcritical bifurcation

$$X' = X(R - X) + \mathcal{O}(X^3)\,.$$

Solution.
We easily see that there is an equilibrium at $x = 1$ for any parameter value r. The linearization shows that it loses its stability at $r = -1$, so this might be the point of the transcritical bifurcation. To transform

$$x' = r \ln x + x - 1 \tag{7.2}$$

into

$$X' = X(R - X) + \mathcal{O}\big(X^3\big) \tag{7.3}$$

we first introduce new coordinates $y = x - 1$ and $R = r + 1$ that shift the bifurcation point to $(0, 0)$. This transforms (7.2) into

$$y' = (R - 1)\ln(y + 1) + y \; . \tag{7.4}$$

Next we expand the logarithm around $y = 0$ which yields

$$y' = (R - 1)[y - \frac{1}{2}y^2 + \mathcal{O}\left(y^3\right)] + y \tag{7.5}$$

$$= Ry - \frac{1}{2}(R - 1)y^2 + \mathcal{O}\left(y^3\right) \; . \tag{7.6}$$

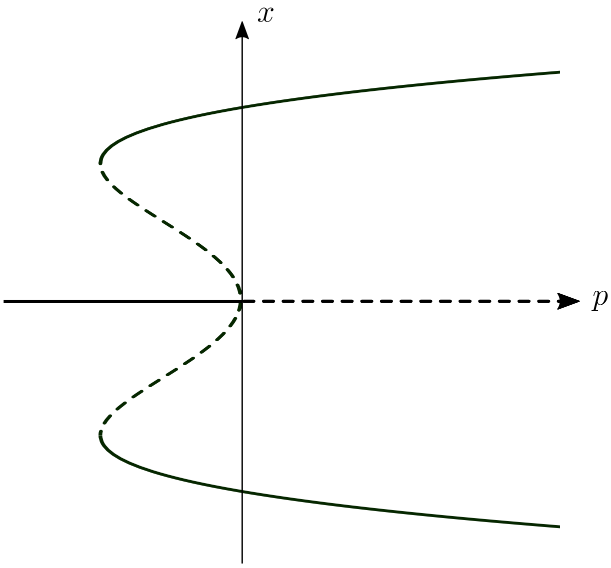
The factor $\frac{1}{2}(R - 1)$ of the quadratic term is the only difference to the desired form (7.3). To get rid of it we perform one further change of coordinates, a scaling of the form $X = ay$. We find that the choice $a = \frac{1}{2}(R - 1)$ indeed brings (7.6) into (7.3). $\square$

Exercise 7.3 (More bifurcations I)**.**
Find all bifurcations of the parametrized system given by

$$x' = -x^5 + x^3 + px$$

and draw its bifurcation diagram.

Solution. We solve the equation $-x^5 + x^3 + px = 0$ to obtain the equilibria. We see that $x = 0$ is an equilibrium for any p. To get a formula for the other equilibria this time it is far easier to solve the remaining equation $-x^4 + x^2 + p = 0$ for p instead of x; that simply yields $p = x^4 - x^2$. Next we calculate $\partial_x f(x, p) = -5x^4 + 3x^2 + p$. Inserting $x = 0$ yields $\partial_x f(0, p) = p$, so the equilibrium at $x = 0$ is stable for $p < 0$ and unstable for $p > 0$. For the other curve of equilibria we plug $p = x^4 - x^2$ into the derivative and obtain $\partial_x f = -5x^4 + 3x^2 + x^4 - x^2 = -4x^4 + 2x^2$. Simple analysis shows that this term is positive for values of x between $-\frac{\sqrt{2}}{2}$ and $\frac{\sqrt{2}}{2}$ (which are the minima of the double well $x^4 - x^2$) and negative outside theses values. Hence the stability looks as follows:



We see that there is a subcritical pitchfork bifurcation at $(x, p) = (0, 0)$. Further there are two fold bifurcations at the minima of the double well, i.e. at $(x, p) = (\pm\frac{\sqrt{2}}{2}, -\frac{1}{4})$. $\square$

Exercise 7.4 (More bifurcations II).
We consider the parametrized map

$$\phi\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}\,, \quad \phi(x,p) = xe^{x+p}$$

(a) Find the fixed points of $\phi(\cdot, p)$ depending on the parameter p. When are they hyperbolic? Determine the stability of the fixed points in the hyperbolic cases.

(b) When the fixed points are non-hyperbolic, bifurcations occur. Classify these bifurcations and draw the bifurcation diagram.

Solution.

(a) Solving the equation $x = xe^{x+p}$ we find the fixed points $x_1 = 0$ and $x_2 = -p$. We have $\partial_x \phi(x,p) = (x+1)e^{x+p}$, so the linearization at $x_1 = 0$ is given by $\partial_x\phi(0,p) = e^p$. Thus x_1 is hyperbolic as long as $p \neq 0$. For the second fixed point $x_2 = -p$ we obtain $\partial_x\phi(-p,p) = 1-p$, so x_2 is hyperbolic if $p \neq 0$ and $p \neq 2$.

In the hyperbolic cases we can determine the stability of the fixed points by the corresponding linearization. Consequently x_1 is stable if $p < 0$ and unstable if $p > 0$. For x_2 we see, that it is stable if $p \in (0,2)$ and unstable if $p \in (-\infty, 0) \cup (2, \infty)$.

(b) Since we have two branches of fixed points which intersect at the origin and exchange their stability we have a transcritical bifurcation at $p = 0$.

As $\partial_x\phi(-2,2) = -1$, we suspect that a flip bifurcation occurs at $p = 2$. We quickly compute

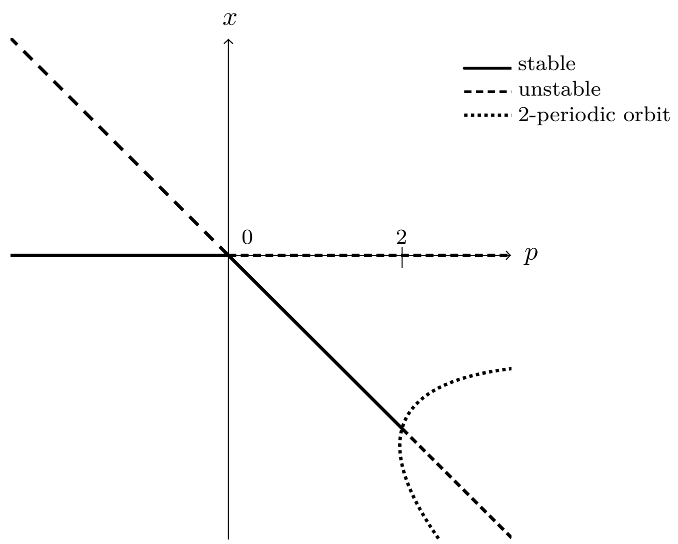
$$\partial_{xx}\phi(x,p) = (x+2)e^{x+p}\,, \partial_{xxx}\phi(x,p) = (x+3)e^{x+p} \;\text{ and }\; \partial_{xp}\phi(x,p) = (x+1)e^{x+p}\,.$$

Thus we have

$$\frac{1}{2}\partial_{xx}\phi(-2,2)^2 + \frac{1}{6}\partial_{xxx}\phi(-2,2) = 0 + \frac{1}{6} \neq 0 \;\text{ and }\; \partial_{xp}\phi(-2,2) = -1 \neq 0\,,$$

so by Theorem 8.13 indeed a flip bifurcation takes place at $(x,p) = (-2,2)$.

The bifurcation diagram looks as follows:



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